

Title

Pneumonia Associated with Invasive and Noninvasive Oxygenation Strategies for Acute Hypoxemic Respiratory Failure in Adults: A Systematic Review and Network Meta-analysis Protocol

Author Affiliations

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Background

Noninvasive oxygenation strategies, such as noninvasive positive pressure ventilation (NPPV) and high-flow nasal cannula (HFNC), have become common treatments for acute respiratory failure (ARF). Compared with invasive mechanical ventilation (IMV), NPPV reduces hospital stay, morbidity, and mortality in patients with ARF¹. Recently, using a systemic review and network meta-analysis approach, Ferreyro BL et al. have reported that treatment with noninvasive oxygenation strategies was associated with a lower risk of death than standard oxygen therapy in adult patients with acute hypoxemic respiratory failure². However, further research is needed to understand the relative benefits and risks of each strategy.

While NPPV reduces the risk of pneumonia associated with IMV, aspiration pneumonia has been described in as many as 5% of NPPV patients³, suggesting a potential risk of pneumonia associated with NPPV. Consistently, Kohlenberg et al. reported that NPPV might increase pneumonia risk compared with no ventilation, according to the nosocomial infection surveillance system database⁴. In contrast, previous investigations have reported a lower incidence of pneumonia with NPPV than with no ventilation⁵. Therefore, the risk of pneumonia associated with NPPV remains controversial.

In addition, noninvasive oxygenation devices, NPPV or HFNC, having more favorable effects for patients with acute hypoxemic respiratory failure have not been thoroughly investigated. Frat et al. reported a lower mortality rate in the group treated with HFNC than the group treated with standard oxygen therapy or NPPV in a randomized clinical trial⁶. However, the differences among the noninvasive oxygenation intervention outcomes remain unclear.

Review question

Which noninvasive oxygenation strategy is best for reducing pneumonia in patients with acute hypoxemic respiratory failure?

Searches

We searched for eligible trials using the following databases: 1) The Cochrane Central Register of Controlled Trials, 2) MEDLINE via PubMed, 3) EMBASE, 4) and Ichushi, a database of Japanese papers. The languages and countries in which the studies were conducted were restricted to English and Japanese.

Type of studies to be included

We will include randomized controlled trials. The languages and countries in

which the studies were conducted were restricted to English and Japanese. Randomized crossover trials, cluster-randomized trials, and quasi-experimental trials were excluded.

Condition or domain being studied

Acute hypoxemic respiratory failure is organ failure that is frequently observed in critically ill patients. While ventilation management by endotracheal intubation is a reliable method of respiratory support, IMV has been associated with various adverse events. Avoiding endotracheal intubation and noninvasive oxygenation strategies such as NPPV or HFNC can reduce severe complications and improve outcomes in respiratory failure management. On the other hand, patients treated with NPPV may have a potential risk of aspiration pneumonia compared with no ventilation. Furthermore, whether noninvasive oxygenation devices, NPPV or HFNC, have more favorable effects for acute hypoxemic respiratory failure patients have not been thoroughly investigated. Few studies have investigated the risk of pneumonia associated with noninvasive oxygenation strategies and compared these two devices in patients with acute hypoxemic respiratory failure.

Type of participants

Inclusion:

We will include patients aged 18 years or older who have acute respiratory failure, defined by the new onset (<7 days) of clinical signs (e.g., tachypnea, increased work of breathing), radiologic signs (unilateral or bilateral chest X-ray opacities), and hypoxemia. Hypoxemia was defined as a P/F ratio below 300 cm H₂O, SaO₂/SpO₂ <94% in room air, PaO₂ <60 mmHg with room air, or <80 mmHg with O₂.

We will utilize studies that include patients treated in the intensive care unit, intermediate care unit, and emergency department.

Exclusion:

- 1) Hypercapnia (e.g., >50 mmHg) in more than 50% of the study's population
- 2) Patients with congestive heart failure, chronic obstructive pulmonary disease, or asthma as the cause of respiratory failure, and more than 50% of the study population
- 3) Post-extubation respiratory failure (>50% population)
- 4) Post-surgical patients (>50% population)
- 5) Post-trauma patients (>50% population)
- 6) The absence of informed consent
- 7) Do-not-resuscitate orders
- 8) Intervention is limited in the emergency department or pre-hospital care

Type of interventions

- 1) Conventional oxygen therapy: low-flow nasal cannula, face mask, and Venturi mask (flow rate is not limited).
- 2) NPPV (type of mask, mode, duration of ventilation, and weaning methods are not limited).
- 3) HFNC (The flow rate and fraction of inspired oxygen are not limited).
- 4) IMV: via endotracheal intubation, with or without a lung-protective strategy.

Type of comparators

All interventions mentioned above can have controls.

Main outcomes

- 1) The incidence rate of pneumonia: ventilator-associated pneumonia (VAP), aspiration pneumonia, and nosocomial pneumonia are not limited.
- 2) The incidence rate of VAP (VAP will be defined in each study)

* Measures of effect

Each outcome and total participants in each group were used to calculate the relative risk (risk ratio).

Additional outcome

None

* Measures of effect

None

Data extraction (selection and coding)

Two of three physicians (HO, SK, SH) will screen the title and abstract at the first screening and the full text at the second screening for relevant studies and independently extract data from included studies into standardized data recording forms. Disagreements will be resolved by discussion with one of three physicians not involved in screening the various studies. We will ask the original author when deemed necessary. If the studies have only the abstract and we cannot assess whether they meet our review criteria, we will contact the authors. If there is a discrepancy between the two reviewers, an agreement will be reached through discussion. Discrepancy checks will be resolved by discussing with the third reviewer as necessary.

We will extract the following study characteristics;

- 1) Methods: study design, total duration of the study period, number and locations of study centers, study setting, withdrawals, and date of the study.
- 2) Participants: N, mean age, age range, sex, severity of condition, diagnostic criteria, inclusion criteria, and exclusion criteria.
- 3) Interventions: intervention method and comparison method.
- 4) Outcomes: specified and collected primary and secondary outcomes, and time points reported.

One review author (MS) will copy the data from the data collection form into the Review Manager (RevMan) 5 file.

Risk of bias (quality) assessment

We will assess the risk of bias of primary outcomes in included studies independently by two of three review authors (HO, SK, SH) using Cochrane Risk of Bias tool 2.0⁷ for the following seven domains: (a) random sequence generation, (b) allocation concealment, (c) blinding of participants and personnel, (d) blinding of outcome assessors, (e) incomplete outcome data, (f) selective outcome reporting, and (g) other sources of bias. Each bias will be graded as either 'low-risk,' 'unclear-risk,' or 'high-risk'. The risk of bias of secondary outcomes in the included studies will be also assessed using Cochrane Risk of Bias tool 2.0. If there is a discrepancy between the two reviewers, agreement will be reached through discussion. Discrepancy checks will be resolved by discussing with the third reviewer as necessary. Each domain will be evaluated in one of three categories: high risk, low risk, and some concerns.

Strategy for data synthesis

Give the planned general approach to be used. Where appropriate, a brief outline of the analytic approach should be provided.

Data synthesis: Forest plots will be used for meta-analysis, and effect size will be expressed as relative risk with 95% confidence interval (CI) for categorical data and weighted mean differences with 95% CI for continuous data. Outcome measures will be pooled using a random effect model to take into account study-specific effects in measures. For all analyses, a two-sided p-value <0.05 will be considered significant. We will synthesize the data using RevMan 5.3 (2014).

Assessment of heterogeneity: Study heterogeneity between trials for each outcome will be assessed by visual inspection of the forest plots and with an I^2 statistic for quantifying inconsistency (RevMan) (I^2 values of 0% to 40%: might not be important;

30% to 60%: may represent moderate heterogeneity; 50% to 90%: may represent substantial heterogeneity; 75% to 100%: considerable heterogeneity). When heterogeneity is identified (I^2 statistic >50%), we will investigate the reason. We will quantify it using the χ^2 test (p -value).

Network meta-analysis will be performed using a frequentist based approach with multivariate random effects meta-analysis. Inconsistency of the network model will be estimated using inconsistency factors and their uncertainty. In addition, ranking plots (rankograms) will be constructed using the probability that a given treatment has the highest event rate for each outcome. The surface under the cumulative ranking curve (SUCRA), which is a simple transformation of the mean rank, will be used to set the treatments' hierarchy. We will perform a meta-analysis using the GRADE Working Group Approach for a network meta-analysis⁸.

Reference

1. Rochwerf B, Brochard L, Elliott MW, et al. Official ERS/ATS clinical practice guidelines: noninvasive ventilation for acute respiratory failure. *Eur Respir J* 2017;50.
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Contact details for further information

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Conflicts of interest

None

Country

Japan